

MARIAN COLLEGE KUTTIKANAM (AUTONOMOUS)
UG Programme (Honors) Degree March 2026 Regular Examinations

Second Semester

252CCRUBC103 Operating Systems

Time: 2 Hours

Maximum Marks: 40

Answer one question from each bunch

BUNCH I

- I.A ✓ Consider the following set of processes with their arrival times and burst times:

Process	Arrival Time	Burst Time
P1	0	7
P2	1	3
P3	2	8
P4	3	5

Draw the Gantt chart and calculate the average waiting time and average turnaround time for each algorithm.

- a) First Come First Serve (FCFS)
- b) Shortest Job First (SJF – Non-preemptive)
- c) Shortest Remaining Time First (Preemptive SJF)
- d) Round Robin (Time Quantum = 1 ms)

(CO2:10 Marks)

OR

- I.B.1 In a time-sharing operating system, multiple processes are submitted by users for execution. Initially, all processes enter the job queue. As memory becomes available, some processes are moved to the ready queue. While executing, a process may request I/O operations and get transferred to a device (waiting) queue. After completing the I/O operation, it returns to the ready queue for further execution. Explain the role of different scheduling queues in this scenario. How does the

movement of processes among these queues help in efficient process management? (CO2:5 Marks)

I.B.2 Consider the following set of processes.

Process	P1	P2	P3	P4	P5
Arrival	0	2	3	6	30
Burst Time	10	12	14	16	05

(CO2:5 Marks)

I.B.(a). Draw a Gantt chart that illustrates the execution of these processes using the preemptive Shortest Job First (SJF) algorithm and the preemptive priority scheduling algorithm. The priority of each process is $P1 = 4$, $P2 = 3$, $P3 = 5$, $P4 = 1$, and $P5 = 1$.

I.B.(b). Find the average waiting time in both methods.

BUNCH II

II.A ✓ There are 4 processes (P0) through (P3) and 3 resource types A (8 instances), B (6 instances), and C (5 instances).

At time (T0):

Process Allocation (A B C) Max (A B C)

P0 1 0 1 3 2 2

P1 2 1 0 4 2 1

P2 1 2 1 2 3 3

P3 1 0 2 3 3 3

Available = (2, 3, 1).

Check whether the system is in a safe state or not using Banker's Algorithm. If it is safe, provide a safe execution sequence.

(CO3:10 Marks)

OR

II.B A system allows only one process at a time to use a shared resource, such as a printer. Multiple processes are trying to access it simultaneously. Using a semaphore S, explain how the operations wait(S) and signal(S) are used to ensure mutual exclusion. What will happen if a process tries to execute wait(S) when $S = 0$?

(CO3:10 Marks)

BUNCH III

- III.A Consider a system with 4 page frames using the FIFO page replacement algorithm.
Given the reference string: 2, 1, 2, 3, 4, 2, 5, 1, 2, 3, 4, 5,
Determine the number of page faults, hit ratio, and miss ratio,
assuming all frames are initially empty. (CO4:10 Marks)

OR

III.B

- III.B.1 Use the given page reference string to explain the following page replacement algorithm for a memory with three frames.
5, 0, 1, 0, 2, 3, 0, 2, 4, 3, 3, 2, 0, 2, 1, 2, 7, 0, 1, 1, 0

(CO4:10 Marks)

- III.B.(a). FIFO
III.B.(b). LRU
III.B.(c). Optimal

BUNCH IV

- IV.A.1 A disk has pending I/O requests at cylinders 98, 183, 37, 122, 14, 124, 65, and 67. The disk head starts at 40. The system administrator considers switching from FCFS to SSTF to reduce seek time. Compare the total head movement in both First Come First Serve (FCFS) and Shortest Seek Time First (SSTF). Which scheduling algorithm is more efficient in this case, and why? (CO5:5 Marks)
- IV.A.2 An enterprise organization requires a file system that supports large volumes and provides encryption and reliability. They are choosing between FAT, NTFS, and EXT. Which file system would best suit their needs, and why? Justify your answer by comparing key features like security, file size support, and reliability. (CO5:5 Marks)

OR

- IV.B.1 In a desktop computer system, multiple hardware components such as the graphics card, hard disk controller, and additional expansion devices, need to communicate efficiently with the CPU and main memory. During a system upgrade, a technician installs a high-performance graphics controller and a storage controller that are connected through a high-speed bus. The technician also explains that

the system uses different controllers and interfaces to manage communication between devices, memory, and the processor. Discuss how components such as the PCI bus, memory controller, disk controller, and expansion bus interface help in coordinating data transfer and improving overall system performance. (CO5:5 Marks)

- IV.B.2 In a database server used by multiple users, a large number of disk I/O requests are generated continuously at different track locations. The disk controller currently uses a scheduling policy in which requests are serviced strictly in the order in which they arrive. During peak working hours, the system experiences a heavy workload and users notice that data retrieval becomes significantly slower. This is because the disk head frequently moves back and forth between distant tracks, leading to increased seek time and reduced disk performance. To improve the efficiency of the system, the administrator plans to implement a more optimized disk scheduling algorithm that can minimize unnecessary head movement and provide faster response to user requests. Suggest a suitable disk scheduling algorithm and justify how it helps in reducing seek time, improving throughput, and enhancing overall system performance. (CO5:5 Marks)

CO1	Describe the structure, types, and services of operating systems.
CO2	Analyze the performance of various process Scheduling Algorithms.
CO3	Appraise various techniques for process synchronization and deadlock handling.
CO4	Analyze the methods employed for memory management in computer systems
CO5	Analyse various disk scheduling and understand file system concepts.